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Key Points:

- A graph neural network for extracting features from the output variables of the numerical weather prediction (NWP) model for rainfall forecasting
- A level-specific training label and a weighted loss function based on the relative frequency of rainfall amount to better training the model
- The proposed method improves the forecast skill at all rainfall intensity levels and can yields more reasonable location and intensity of rainfall

Supporting Information:

Supporting Information may be found in the online version of this article.

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A Structured Graph Neural Network for Improving the Numerical Weather Prediction of Rainfall

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Abstract The current challenges of numerical weather prediction (NWP) of rainfall mainly stem from the complex and multiscale nature of rainfall. In recent years, as observation capability improved worldwide, there has been an increased feasibility to use data-driven models to enhance forecasting performance with rainfall observation. Compared to traditional statistical and machine learning models, deep-learning models show considerable promise in capturing the spatial-temporal features of weather processes from multiple predictors, but the convolution-based feature extractor is suboptimal due to the linear nature of convolution kernels. In this study, a multilevel forecasting model is proposed to forecast each rainfall level, in which each submodel adopts a graph neural network for feature extraction. Spatial and temporal propagation functions based on grid structure are designed to explicitly represent feature fusion and propagation of multiple predictors across multiple scales. On model training, a weight setting strategy that balances the impact of samples with different rainfall values on the total training loss is proposed, and a soft classification label is designed to convert observed rainfall into the probability of rainfall above each threshold. The proposed model was trained and validated on NWP data provided by European Center for Medium-Range Weather Forecast, and results show significant improvement over the NWP in terms of threat score (TS) and Heidke Skill Score (HSS) scores. Analysis of the forecast results for two typical rainfall processes also illustrates that the proposed method can predict rainfall with more reasonable location and intensity.

Plain Language Summary Rainfall forecast plays an important role in human's daily life, and numerical weather prediction (NWP) has helped to greatly improve the performance of rainfall forecast. However, accurate forecasting is still challenging due to the complex characteristics of rainfall and uncertainty of NWP models. In recent years, there has been accumulated large amount of high-precision rainfall observation data around the world. It becomes possible to enhance the forecasting performance with these data. However, the traditional statistical models either ignore the weather background information or struggle to capture the complex interactions between different variables produced by the NWP model. To handle these shortages, a neural network model based on grid structure is proposed, which can extract multiscale features from the output variables of the NWP model through an information propagation mechanism. Meanwhile, a multilevel forecasting strategy is proposed for better handling different rainfall levels along with different uncertainties, from light rain to heavy downpours. The proposed model was applied to a real NWP data for the central part of East China, and the results illustrate significant improvement in terms of common forecast skills and intuitive feeling.

1. Introduction

Rainfall has long been one of the most important weather phenomena that affect human life. Accurate prediction of rainfall is of great significance in many aspects, including transportation, agriculture, disaster early warning, etc. In recent years, numerical weather prediction (NWP) has undergone significant advancements (Bauer et al., 2015), leading to notable improvements in the spatial and temporal range, as well as the accuracy of rainfall forecasting. As a result, NWP has undoubtedly now become the cornerstone of operational rainfall forecast. However, as the distribution of rainfall is characterized as multiscale, nonlinear, and non-Gaussian (Koutsoyianis, 2004), it is still difficult to make a precise forecast of rainfall. The main problem is that current operational NWP models have large uncertainty at meso-microscale due to the following reasons. First, the spatial resolution of the major operational global NWP models is 10–20 km, which cannot explicitly resolve many of the relevant processes below the convective scale. These processes are usually approximated by subgrid parameterization

(Bauer et al., 2015), that is, they are represented as functions with respect to model resolved variables. The parameterizations directly related to precipitation include convection parameterization (Wilcox & Donner, 2007) and cloud microphysics parameterization (Hong et al., 2004; Hong & Lim, 2006; Wicker & Wilhelmson, 1995). However, parameterization is a simplified approximate description of the subgrid process, which usually smooths out the details of the small-scale processes. For some specific weather processes, parameterization shows large uncertainty. Second, though some high-resolution NWP models can directly resolve the process of the convective scale (Baldauf et al., 2011; Clark et al., 2016; Tao et al., 2012), they require more accurate analysis data for initialization and higher computational capability. These factors make the high-resolution model not widely applied in operation. In addition, as the initial field error and numerical approximation error always exist, and the atmosphere is a highly nonlinear chaotic system, the uncertainty of NWP always exists (Boeing, 2016; Slingo & Palmer, 2011).

With the enhancement of observation capability, a large number of high-resolution rainfall data have been accumulated around the world (Li et al., 2022; Petersen et al., 2002; Zhou et al., 2008). Some statistical postprocessing methods have been developed to correct the bias of rainfall with these observation data. Model output statistics (MOS) (Glahn & Lowry, 1972; Wilson & Vallée, 2002), quantile mapping (QM) (Cannon, et al., 2015; Déqué, 2007), and frequency matching (FM) (Zhu & Luo, 2015) are the most commonly used bias correcting methods in operations. For instance, MOS employs linear regression to establish the relationship between the outputted variables from NWP model at a given grid point and the observed rainfall at the nearest station. This relationship is then used to forecast rainfall either at the station or the grid point. Though linear model has the characteristics of simplicity and strong interpretability, MOS ignores the spatial relation between variables due to it only considers the relationship between variables at a single point. Assuming that the forecasting bias has a certain consistency in a short period of time, QM and FM utilize the difference of frequency between recently observed and forecasted rainfall at a certain rainfall intensity range to correct distribution of forecasted rainfall, so that it is closer to the observed distribution. However, this statistical distribution is calculated on all grid points in the area interested with discard of the location information. And the correction of each grid point is performed independently, which ignores the correlation of neighboring grid points. Moreover, as the correction only considers the single variable of rainfall, the information of synoptic background is also discarded.

In fact, many statistical correction methods are inherently data-driven and can be effectively implemented using machine learning techniques (Ben-Bouallègue et al., 2023; Vannitsem et al., 2021). In recent years, with the improvement of computing power and the developing of advanced optimization methods, deep learning (LeCun et al., 2015), as a branch of machine learning, has achieved great success. These models can automatically learn complex nonlinear relationships from a large amount of data, which has been utilized to improve the performance of NWP models, including improving the parameterization (Brenowitz & Bretherton, 2019; Gentile et al., 2018; Krasnopolsky et al., 2013; Ma & Tan, 2010) and correcting the output variables (Grönquist et al., 2021; Han et al., 2021; Peter et al., 2021). Replacing part of parameterization models with deep-learning models can improve the accuracy or computational efficiency of parameterization (Nonnenmacher & Greenberg, 2021). However, parameterization usually only uses variables at the same grid point of the same time step, which could lead to oversimplification. Thus, improving the parameterization has limited benefits to complex phenomenon such as precipitation. Besides that, correction based only on the variable of rainfall could not take full advantage of deep learning on fitting high-dimensional nonlinear relations. Most deep-learning models use multiple variables as input, and try to learn a map from the NWP model's output variables (predictors) to the predicted rainfall field. There are mainly two ways of modeling. One is using the predictors in a local area centered on a grid point as input, to output a rainfall value of the same grid point, that is, the (discretization) area-to-point mapping (Liu et al., 2022; Pan et al., 2019). The other is using the predictors in the whole area as input, to output the predicted rainfall field of the same area, that is, area-to-area mapping (Adewoyin et al., 2021; Hess & Boers, 2022). In addition, some studies have shown that using predictors of multiple time steps achieves better results than using those of only a single time step (Liu et al., 2022). However, the multiscale character of rainfall requires the model to be capable of representing multiscale features. The most deep-learning models mainly use existing convolutional neural networks (CNN) or recurrent neural networks (RNN), and treat the NWP data as multichannel images. Although CNN can extract multiscale features by multiple times of convolution, but the linear convolution shows limited capability of representing complex interactions between multiple variables. On the other hand, a large forecast area can have multiple types of rainfall with differentiated driving factors, leading to varying degrees of uncertainty across different parts of the rainfall. It is crucial to recognize and account for these categorical

distinctions within the forecast area. Based on the above two concerns, a structured graph neural network (GNN) is proposed to extract spatial-temporal features from multiple predictors through mixing and propagating information along neighboring grid points. And then, multiscale spatial and temporal features can be extracted by conducting the information propagation multiple times. Furthermore, a multilevel forecasting strategy using submodels for different rainfall levels is proposed, and the forecasts of submodels are merged to get the whole forecast result, which can also be applied to any deep-learning model. For each submodel, the structured GNN is employed as the feature extractor, which can be trained by labels designed for the corresponding thresholds. By combining the multilevel forecasting strategy with optimizing of the submodels, the forecasting skills can be improved systematically. It should be noticed that, the proposed model is not intended to replace NWP with a fully data-driven surrogate model. Instead, we use NWP forecasted variables to enhance the prediction of a different variable, namely rainfall. This approach also differs from conventional statistical postprocessing, which primarily aims to refine the original model's forecast based on historical observations.

On the aspect of model training, the sample imbalance is one of the main problems for training the deep-learning-based rainfall prediction models. Some statistical analysis on observed rainfall data shows that, the frequency of rainfall decreases exponentially with the increasing of rainfall amount (Dan'azumi et al., 2010; Kiely & Ivanova, 1999). If using the conventional MSE or MAE as loss functions, predicting errors of samples with large rainfall would be smoothed out. To alleviate this problem, a weighted MAE loss function is proposed. First, the probability density of rainfall amount is estimated with the training rainfall data, which can be used as an indicator of the aggregation at each rainfall amount. Then, a weighting strategy based on the probability density of rainfall amount is proposed to balance the influence of rainfall samples with different rainfall amount.

2. Data

The NWP data used in this paper are the forecasts of the integrated forecasting system (IFS) (Owens & Hewson, 2018) developed by the European Center for Medium-range Weather Forecasts (ECMWF). The high-resolution forecast (HRES) of IFS provides deterministic forecasts of nine variables at 19 vertical layers and 33 single layer variables with spatial resolution of 0.125° and temporal resolution of 3 hr. The area is specialized to the central part of East China (28° – 34.5° N, 115° – 122° E), where the Shanghai Central Meteorological Observatory provides multiple forecast products based on the forecasts of IFS. The forecast data cover the time range from January 2017 to June 2023, with two forecasts made each day (starting at 00:00 and 12:00, UTC). After filtering out examples with completely clear days or invalid data, a total of 2,589 forecasts were retained. For training and validation, we randomly selected forecasts from January 2017 to December 2021, resulting in 1,500 forecasts for the training set and 460 for the validation set. The testing set consisted of 626 forecasts from January 2022 to June 2023.

The rainfall observation data are the real-time observations of automated rain-gauge stations provided by the China Meteorological Administration. The data is transformed to ground truths at grid points via nearest neighbor interpolation. The precision of rain-gauge data is 0.1 mm. Figure 1 shows the research area and the locations of the rain gauges used for training and testing. The average distance between neighboring rain gauges is about 8.5 km, this density is enough for the deep-learning model to learn the rainfall distribution at the grid-scale.

To investigate the relative frequency for each rainfall amount from the historical data in the study area, a grid invariant statistical analysis was performed. To do so, we estimated the probability density function of rainfall value R (i.e., $\text{PDF}(R)$) using the rain-gauge data with the equation given below:

$$\text{PDF}(R) \approx \frac{\text{Freq}(R - \Delta R \leq R < R + \Delta R)}{2 * \Delta R} \quad (1)$$

It means the frequency of rainfall in a small value interval centered at R dividing the interval length. When the interval length goes to 0, we get the exact definition of PDF.

As the precision of the rain-gauge data is 0.1 mm, ΔR should not be too small. And the number of samples decreases significantly with the increase of rainfall value. Therefore, a larger ΔR should be used for numerical stability. In this paper, ΔR was set as 0.1, 0.2, and 0.5 mm for $0 \text{ mm} < R < 3 \text{ mm}$, $3 \text{ mm} \leq R < 10 \text{ mm}$, and $R > 10 \text{ mm}$, respectively. The PDF of $R = 0 \text{ mm}$ was not calculated, as all situations of nonrainfall corresponds to one value, that is, 0, which makes the PDF at 0 extremely large. Thus, the value of 0 should correspond to the

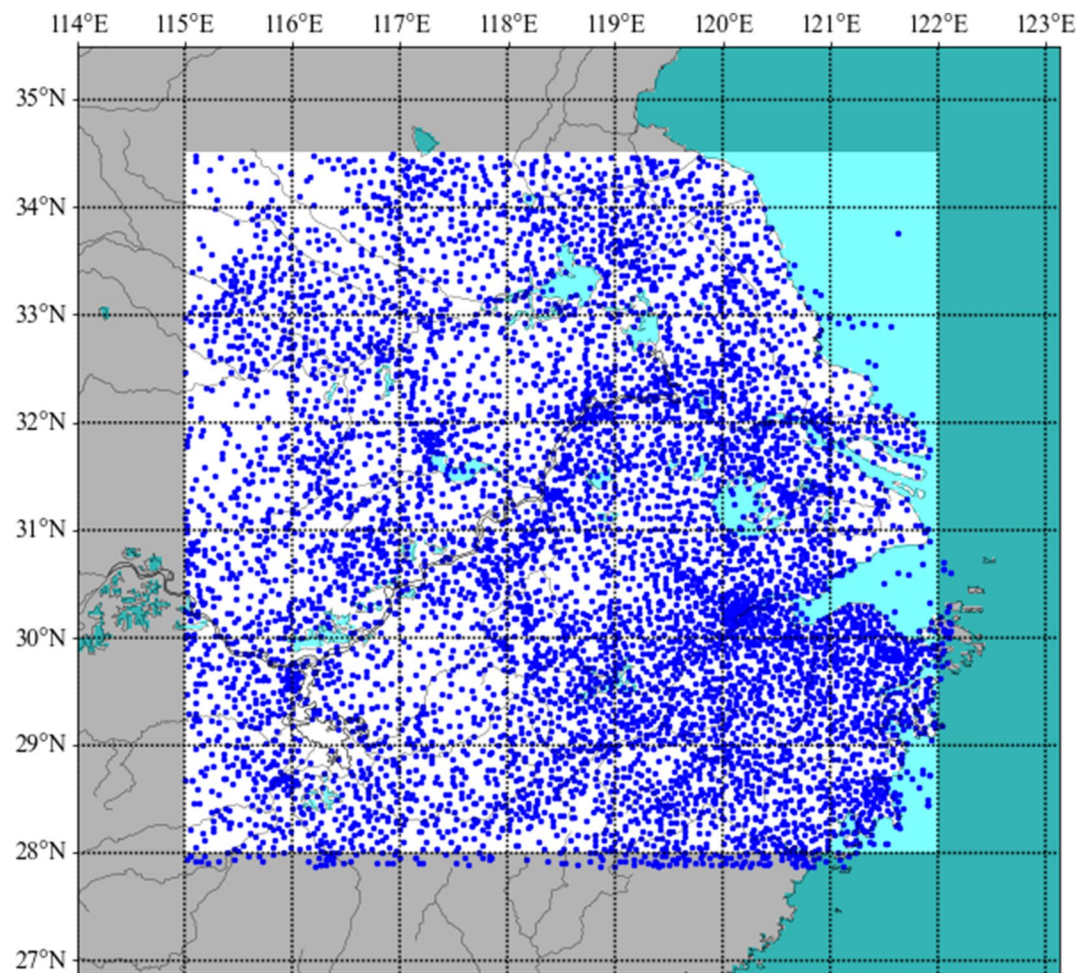


Figure 1. The study area and the locations of automatic rain-gauge stations (blue points) used. To provide more adequate information of the synoptic background, the input area is larger than the study area, with extension of 1° to the east, west, north, and south, respectively.

interval $(-\infty, 0]$ mathematically. Figure 2 shows the curve of $\text{PDF}(R)$, where the red line is a reference line of $y = 10^{-R/10}$. It can be seen that the PDF decreases very fast with the increase of rainfall value when $R < 5$ mm. When $R > 10$ mm, PDF still decreases exponentially, but slower than $10^{-R/10}$.

The samples with 0 mm (nonrainfall) account for about 83.9% of the whole samples, while samples with $R > 10$ mm only account for about 1.2%. This extreme imbalance of samples on different rain levels indicates the necessity for the data-driven models to emphasize more on the infrequent large valued samples. And this statistical analysis is useful in guiding the model training more precisely, which will be further discussed in Section 3.2.2.

3. Methodology

3.1. The Forecast Model

3.1.1. The Overall Framework

To handle the rainfall at different intensity levels, a multilevel modeling strategy is proposed, in which each submodel predicts the rainfall area where values exceed the corresponding threshold. The final forecast is obtained by merging the outputs of all submodels. The overall framework of the forecast model is shown in Figure 3.

3.1.2. Selection of the Predictors

A comprehensive rainfall forecast needs to consider the dynamics, thermodynamics, humidity, and atmospheric stability factors collectively. While introducing more factors will increase the modeling complexity and the

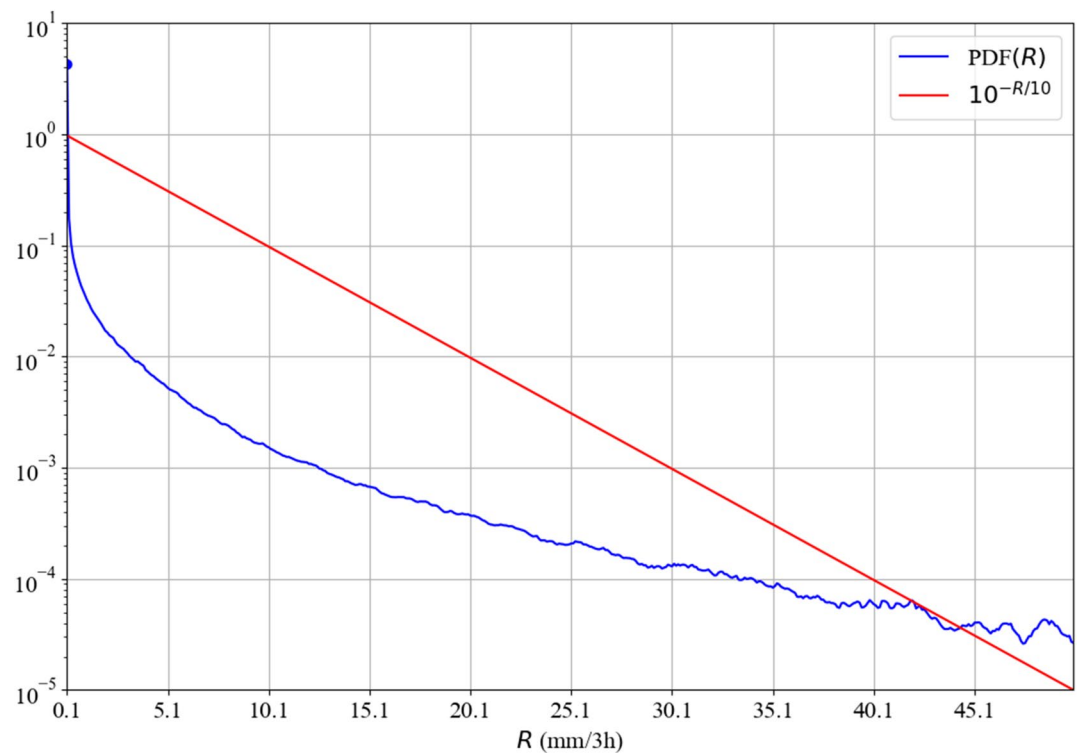


Figure 2. The curve of $PDF(R)$ based on the statistic on all the rain-gauge observations in the research area.

training difficulty, nine variables at five typical barometric levels (200, 500, 700, 850, 925 hPa) and 13 surface variables were chosen as candidate predictors, with consideration of model training efficiency and human forecasters' experience integrally. Several derived variables, such as the K index and the potential pseudoequivalent temperature, were calculated as new predictors. Adding the 13 surfaces variables, there are totally 60 candidate predictors, which are listed in Table S1 in Supporting Information S1.

According to the commonly used standard of operational 12-hr rainfall-level division given in Table S2 in Supporting Information S1, we divided 3-hr rainfall into four levels, of which the lower bounds were 0.1, 3, 10, and 20 mm. And rainfall smaller than 0.1 mm was treated as clear day. Thus, there were four submodels needed based on this division. As the rainfall forecasted at time step t is the accumulated precipitation from time step $t - 1$ to time step t , it is more reasonable to use the difference and the mean of the predictor, i.e., $e_t - e_{t-1}$ and $(e_t + e_{t-1})/2$, as the new predictors. By doing this, we got the expanded set of predictors $\mathbf{P}$. For each submodel at each rainfall level, a level-specific subset of predictors was selected in a way given below:

First, all the predictors were divided into five categories based on their physical meaning, which are direct, dynamic, thermal, humidity, and stability, as shown in Table S1 in Supporting Information S1.

Second, the correlation of the observed rainfall above threshold l and each predictor is calculated, and Pearson correlation coefficient (PCC) was used in this paper.

Finally, the predictors that have relative higher absolute values of PCC in each category are selected.

The computation results of PCCs in the training set indicate that, for thresholds of 0.1 and 3 mm, ranks of PCCs are similar. The same is true for thresholds of 10 and 20 mm. Thus, the same subset of predictors was selected for thresholds of 0.1 and 3 mm. And for thresholds of 10 and 20 mm, the subsets of predictors were also identical. The predictors for each threshold are shown in Table 1. The direct predictors were not used, despite their higher PCCs in certain cases. This is because IFS derives precipitation through parameterization, which inherently introduce additional errors. Therefore, we only utilized more reliable fundamental variables and their associated derivatives to avoid potential adverse effects of IFS derived precipitation in certain cases.

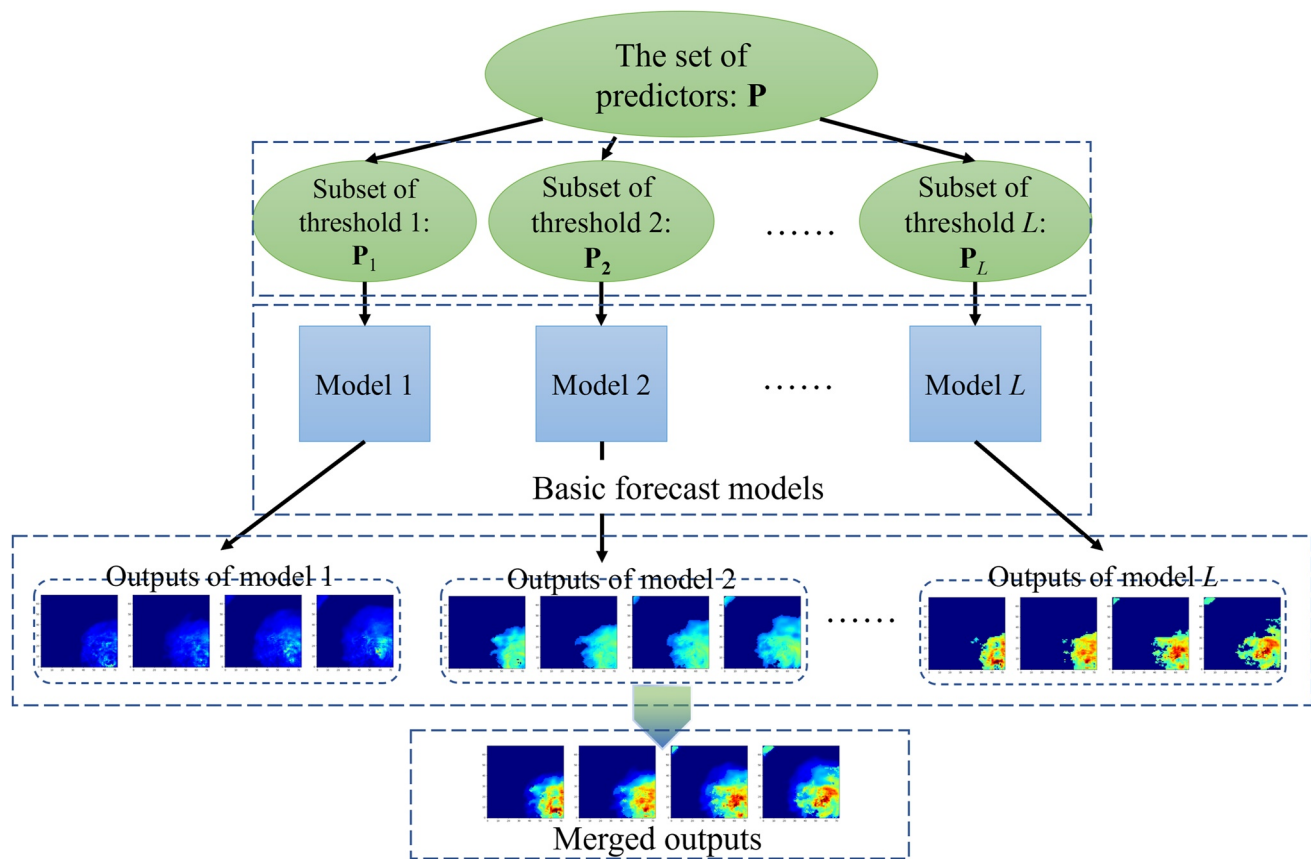


Figure 3. The overall framework of the multilevel forecast model. The model consists of three parts. The first part is the predictor selection module, which first constructs the set of candidate predictors from the output variables of the numerical weather prediction (NWP) model, then selects the predictors for each submodel at each rainfall threshold. The second part contains the basic forecast models at all rainfall threshold. Each submodel receives the level-specific predictors, and outputs the corresponding level forecast. The third part is the multilevel merging module, which combines the outputs of submodels at all levels to generate the final forecast.

As the range of value of each predictor varies greatly, it is necessary to normalize the value of predictors before inputting the model. The normalization was performed as follows:

$$\tilde{e} = \frac{e - \text{mean}(E)}{\text{std}(E)}$$

where e is the value of a specific predictor at an example point in the training set. E is the set of values of the same predictor at all example points.

3.1.3. The Forecast Submodel Based on the Structured Graph Neural Network

For each submodel, the inputs are the normalized predictor of multiple successive time steps, and the outputs are the level-specific rainfall forecast results of the same time steps. As the “future” cannot affect the “present,” to predict the accumulated rainfall from time step $t - 1$ to t , only the factors before t can be considered. Therefore, it is reasonable to construct the model in an autoregressive way, that is, predicting future values based on past values (Connor et al., 1994; Lewis & Reinsel, 1985). The detail of the autoregressive model structure is shown by Figure S1 in Supporting Information S1.

The key to the performance of the model is designing of the feature extractor of each time step, that is, how to represent the spatial-temporal features of the predictors and the joint effect of multiple predictors. Considering the multiscale characteristics of precipitation, the rainfall at a certain place is not only driven by relevant factors in the local area, but may also be controlled by the synoptic background with a larger spatial and temporal scale. To express the effect at multiple spatial and temporal scales, a structured GNN is proposed to explicitly represent the information propagating between grid points. The data are first represented in the form of regular grid data, all

Table 1
The Predictors Selected for Each Threshold

$(e_t - e_{t-1})/2$	Category	$e_t - e_{t-1}$	Category
(a) 0.1 and 3 mm			
Vertical velocity (500, 700, 850 hPa)	Dynamic	Vertical velocity (all five layers)	Dynamic
V component of wind (500, 700, 850 hPa)		Potential vorticity (200, 500 hPa)	
U component of wind (500, 700, 850 hPa)		Divergence (500, 700 hPa)	
Geopotential height (200, 500, 850, 925 hPa)	Humidity	2-m temperature	Thermal
Relative humidity (all five layers)		Maximum temperature at 2 m in the last 3 hr	
Specific humidity (all five layers)		Minimum temperature at 2 m in the last 3 hr	Humidity
Total column water		Total column water	
Total cloud cover	Stability	Total cloud cover	Stability
Low cloud cover		K index	
K index			
(b) 10 and 20 mm			
Vertical velocity (850, 925 hPa)	Dynamic	Vertical velocity (850, 925 hPa)	Dynamic
V component of wind (200, 500 hPa)		V component of wind (200, 500 hPa)	
U component of wind (200, 500 hPa)		U component of wind (200, 500 hPa)	
Potential vorticity (925 hPa)	Thermal	Temperature (all five layers)	Thermal
Geopotential height (200, 500, 925 hPa)		2-m temperature	
Temperature (all five layers)		2-m dewpoint temperature	
2-m dewpoint temperature		potential pseudoequivalent temperature (500, 925 hPa)	
2-m temperature	Humidity	Relative humidity (500, 700, 850 hPa)	Humidity
potential pseudoequivalent temperature (500, 925 hPa)		Specific humidity (850, 925 hPa)	
Relative humidity (500, 700,850 hPa)		K index	
Total column water	Stability		Stability
K index			

the data samples are distributed on the vertices. Spatial aggregation functions based on this grid structure are then implemented to realize information transformation between grid points. The information can be spread to a larger extent by multiple times of transformation. The basic form of the spatial aggregation function is given below:

$$h_t(i, s+1) = \sum_{j \in N_i} f_s((i, s), e_t(j)) \quad (2)$$

where $h_t(i, s)$ is the high-dimensional hidden state of grid point i and scale s ($s = 1, 2, \dots, S$) at time step t , $e_t(i)$ is the vector of predictors of grid point i at time step t , f_s is the aggregation function of scale s , which is used to transform neighboring source node and edge features into a “message” to the target node. N_i is the set of neighboring grid points directly connected to grid point i (including itself). The graphic structure used by the model is shown in Figure 4.

Figure 5 illustrates the forecast submodel based on the information fusion mechanism described by Equation 2. The transferring of hidden state of each node (grid point) from time step $t-1$ to t is similar to Equation 2 and given below:

$$h_t(i, 0) = \sum_{j \in N_i} f_0(h_{t-1}(j, S), e_{t-1}(j)) \quad (3)$$

This type of GNN has a compact structure, as the aggregation function is shared for all the time steps. Such compact structure is supported by a common sense that the basic principle of a physical process is time invariant. In the sense of machine learning, compact structure can greatly reduce the number of parameters and the risk of

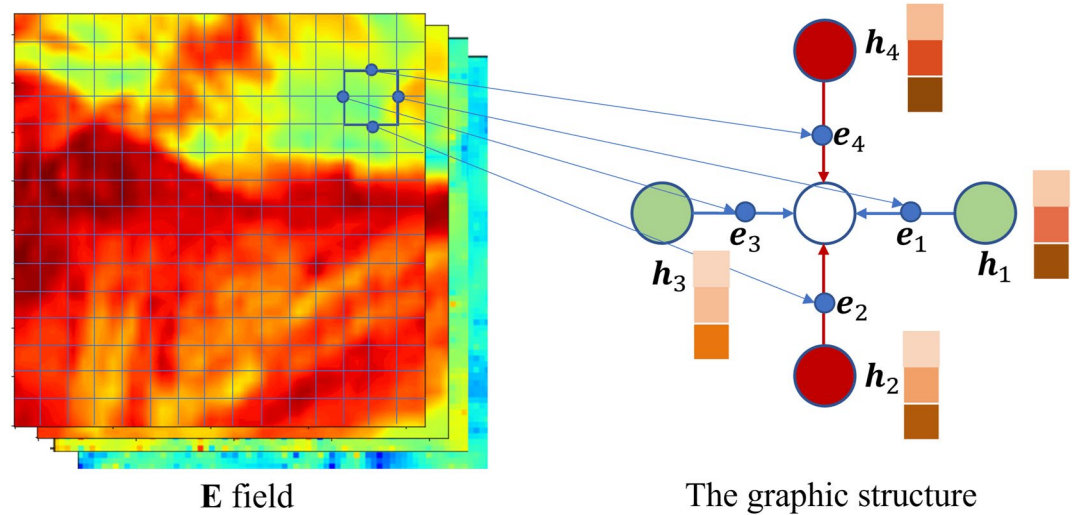


Figure 4. The graphic structure used by the model. Each grid point is treated as a node, and collected with an edge to the four neighboring nodes from left, right, up and down, respectively. The vector of predictors, e , of the neighboring node is treated as edge features to participate in information transformation.

overfitting. By designing the aggregation functions, different types of information transformation can be realized. There are two types of aggregation function proposed, which are the spatial propagating function f_{prop} and the multipredictor mixing function f_{mix} . And we name this model as *Gridded information propagation and mixing network* (GIPMN) accordingly.

The form of f_{prop} is given by Equation 4:

$$f_{\text{prop},s} = \text{ReLU}(\mathbf{W}_{\text{prop}}(s)\mathbf{h}_{\text{prop},t}(j,s)) \odot \mathbf{e}_t(j) \quad (4)$$

where “ $\odot$ ” represents point-wise production. The size of weight matrix $\mathbf{W}_{\text{prop}}(s)$ is $L_e \times L_{h_{\text{prop}}}$, and L_e is the length of the vector of multipredictor, $L_{h_{\text{prop}}}$ is the length of the vector of hidden states $\mathbf{h}_{\text{prop},t}$, which are set manually.

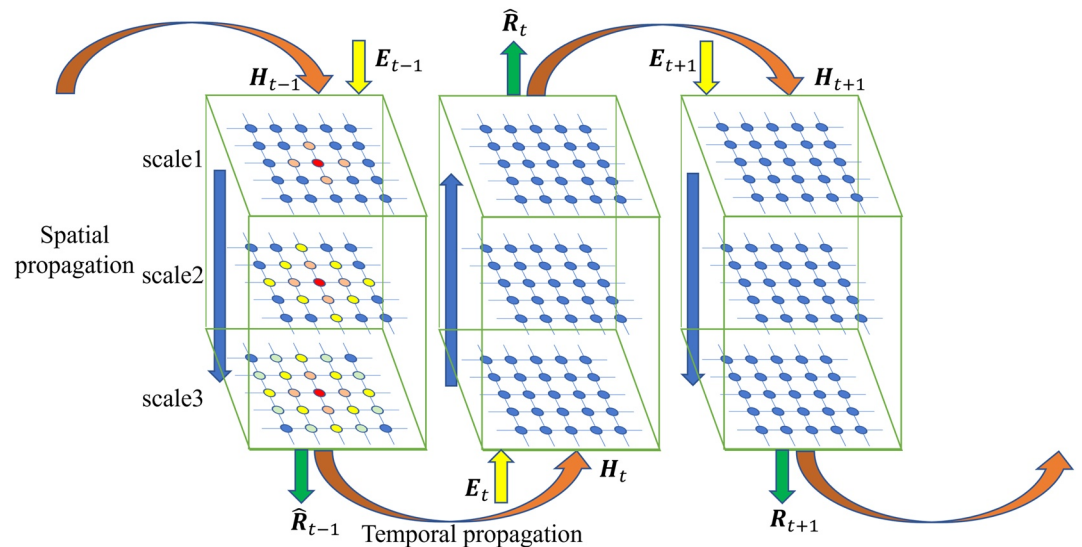


Figure 5. Illustration of the forecast submodel based on graph neural network (GNN). The box in each time step is a feature extractor. In vertical direction, information spreads by multiple times of spatial aggregation, with each time involving four neighboring nodes. The “scale s ” means the spatial extent of information diffusing from a node after s steps of spatial aggregation, as shown by the color changes in the nodes. In horizontal direction, the fused information passes forward in time by transferring the hidden state of each node across adjacent time steps.

$\text{ReLU}(x) = \max(0, x)$ (Nair & Hinton, 2010). The meaning of Equation 4 is that, each predictor multiplies a weight whose value is nonnegative. And the weight is a function of the hidden states $\mathbf{h}_{\text{prop},t}$. The update of the hidden states is done by the aggregation function (Equation 1):

$$\mathbf{h}_{\text{prop},t}^i(s+1) = \sum_{j \in N_i} f_{\text{prop},s}(\mathbf{h}_{\text{prop},t}^j(s), \mathbf{e}_t(j)) \quad (5)$$

The operation described by Equation 5 is similar to convolution, i.e., each predictor passes information independently. The only difference is that, the convolution weight is a function of hidden states of the previous spatial scale, and the value is constrained in $[0, +\infty)$. By setting multiple f_{prop} , i.e., $F_{\text{prop}} = \{f_{\text{prop}}^p | p = 1, 2, \dots, P\}$, different types of spatial feature of predictors can be learned. In this way, P different hidden states $\mathbf{H}_{\text{prop}} = \{\mathbf{h}_{\text{prop},t}^p | p = 1, 2, \dots, P\}$ are derived.

The multipredictor mixing function f_{mix} is expressed in the following form:

$$f_{\text{mix},s} = \sigma(\mathbf{W}_{\text{mix}}(s)[\mathbf{h}_{\text{mix},t}(j, s), \mathbf{e}_t(j)]) \quad (6)$$

where the size of weight matrix $\mathbf{W}_{\text{mix}}(s)$ is $L_{\text{hmix}} \times (L_{\text{hmix}} + L_e)$. And L_{hmix} is the length of the vector of hidden states, which is also set manually. σ is a nonlinear transform function. The aim of Equation 6 is to fuse all the components of multipredictor $\mathbf{e}_t$ and hidden states $\mathbf{h}_{\text{mix},t}$ with the weight matrix $\mathbf{h}_{\text{mix},t}$.

The update of the hidden states is also done by the aggregation function (Equation 1):

$$\mathbf{h}_{\text{mix},t}(i, s+1) = \sum_{j \in N_i} f_{\text{mix},s}(\mathbf{h}_{\text{mix},t}(j, s), \mathbf{e}_t(j)) \quad (7)$$

By multiple times of aggregation with both f_{prop} and f_{mix} , the multiscale merged features, i.e., $\mathbf{h}_{\text{prop},t}^p(i, S)$ and $\mathbf{h}_{\text{mix},t}(i, S)$ are derived. The concatenated feature vector $\mathbf{h}_t(i) = [\mathbf{h}_{\text{prop},t}(i, S), \mathbf{h}_{\text{prop},t}^1(i, S), \mathbf{h}_{\text{prop},t}^2(i, S), \dots, \mathbf{h}_{\text{prop},t}^P(i, S), \mathbf{h}_{\text{mix},t}(i, S)]$ is then inputted into the following mapping layer to get the predicted rainfall field of current time step:

$$\hat{\mathbf{y}}_t(i) = \text{ReLU}(\mathbf{W}_{R,t} \mathbf{h}_t(i)) \quad (8)$$

where the size of $\mathbf{W}_{R,t}$ is $1 \times (L_{\text{hmix}} + P \times L_{\text{hprop}})$, the usage of ReLU is ensure nonnegativity of rainfall value. It should be noticed that, $\hat{\mathbf{y}}_t$ is not necessarily equivalent to the predicted rainfall. Because to make the training more efficiently, the observed rainfall R is usually converted to label y . The fitting target of $\hat{\mathbf{y}}_t$ is then y , not R . The predicted rainfall $\hat{\mathbf{R}}_t$ is derived by inverse transformation from $\hat{\mathbf{y}}_t$. The way of constructing the training label is explicitly described in Section 3.2.1.

3.1.4. Merging of the Submodels' Forecasts

For each rainfall level, a submodel is constructed by using GIPMN described in Section 3.1.3. Each submodel predicts the level-specific rainfall field, and the complete forecast is generated by multilevel merging, of which the main steps are given below:

First, for each threshold l , determine the initial rainfall area above this threshold according to the submodel's output $\hat{\mathbf{R}}_{t,l}$:

$$\text{mask}(l) = \left\{ i \mid \hat{\mathbf{R}}_{t,l} \geq th_l \right\}$$

where, th_l is the lower bound of level l .

Then, correct the rainfall area according to the principle that the area with rainfall above a higher threshold must be included by that above a lower threshold:

$$\text{mask}_{\text{cal}}(l+1) = \text{mask}(l+1) \cap \text{mask}(l)$$

$$\text{mask}_{\text{cal}}(1) = \text{mask}(1)$$

where $\text{mask}_{\text{cal}}(l)$ is the calibrated rainfall area above threshold l . The merging process is illustrated in Figure 6.

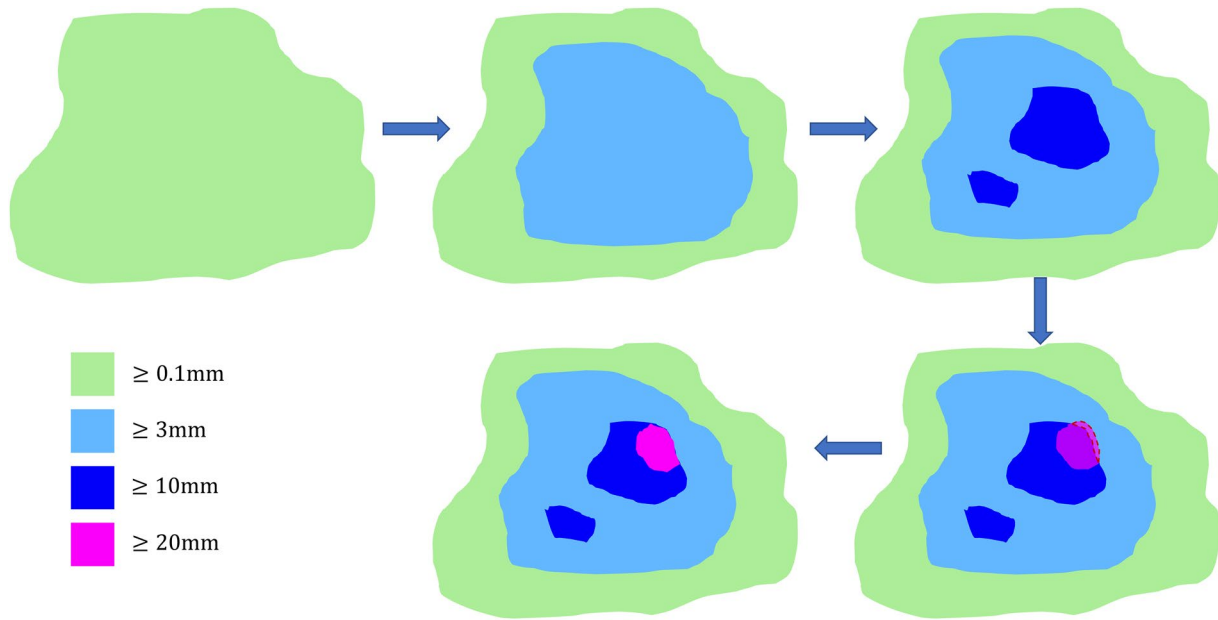


Figure 6. Illustration of the merging process of multilevel outputs. The area with boundary marked by dotted line is removed with the principle described above.

3.2. Model Training

3.2.1. Rainfall-Level-Specified Training Label

For the submodel of each level, a classifier can be used to judge whether the rainfall at each grid point is larger than the threshold (lower bound) of that level. A common rule is that outputting “0” indicates smaller than the threshold, and “1” indicates larger than the threshold. However, rainfall is a continuous variable, rainfalls in the vicinity of a certain level correspond to several typical weather processes. Therefore, the mapping from the input predictors to the predicted rainfall should be a continuous function within this range, and hard classification labels (0 and 1) will destroy this continuity. Considering that the atmospheric state simulated by a NWP model has a certain degree of uncertainty, a soft classification label (except for the submodel of 0.1 mm) with a segmented function is introduced to reflect the continuous change of uncertainty with rainfall value and defined as follows:

$$y = \begin{cases} \frac{0.1}{th_l - \delta R_l} R & , R < th_l - \delta R_l \\ \text{sigmoid}(k_l(R - th_l)) & , th_l - \delta R_l \leq R \leq th_l + \delta R_l \\ \frac{0.1}{th_l - \delta R_l} (R - (th_l + \delta R_l)) + 0.9 & , R > th_l + \delta R_l \end{cases} \quad (9)$$

where $k_l = \ln 9 / \delta R_l$, δR_l represents the “width” of each level, which was set to be 1 mm, 3 and 10 mm for $th_l = 3$ mm, 10 and 20 mm, respectively, in this study. For rainfall around the threshold of a level, slight difference in initial-ization will lead to two different forecast results, that is, one is above the threshold, the other is the opposite. It means that, the uncertainty of whether predicted rainfall being larger or smaller than the threshold is maximized at the threshold, i.e., $P(R \geq th_l) = P(R < th_l) = 0.5$. The uncertainty is reduced when away from the threshold.

The transformation defined by Equation 9 maps the values within the range of $[th_l - \delta R_l, th_l + \delta R_l]$ to the range of $[0.1, 0.9]$. Therefore, within this range, the model’s output is similar to the “probability” that the predicted rainfall $\hat{R} \geq th_l$. When $R = th_l$, $y = 0.5$, indicating that the probability of rainfall being larger or smaller than the threshold is equal. When $R > th_l$, the probability of rainfall being larger than the threshold increases. A linear mapping with a small slope is used to “compress” the label values outside the range of $[th_l - \delta R_l, th_l + \delta R_l]$. This setting allows the model to focus more on learning the characteristics of the samples with rainfall near th_l . Figure 7 shows the curve of the mapping from observations to labels.

For the 0.1 mm level, since the observed rainfall are discontinuous between 0 and 0.1 mm and the two values represent “nonrainy” and “rainy,” respectively. The submodel of 0.1 mm can be regarded as a binary classifier. In

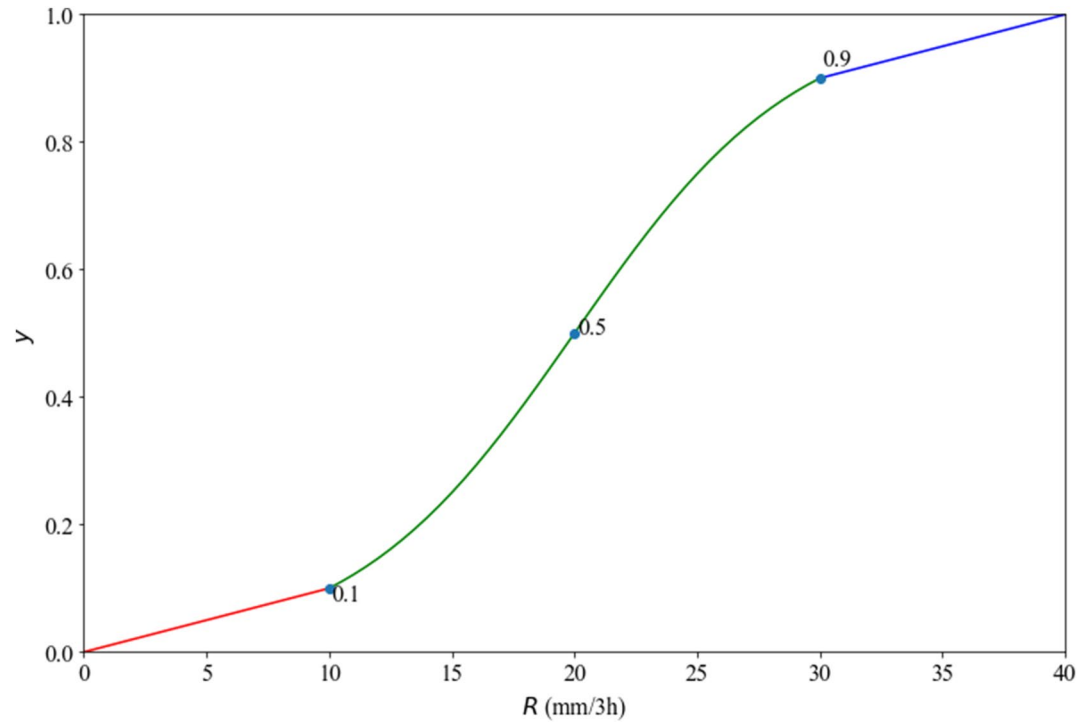


Figure 7. Illustration of the mapping method of rainfall amount to training label for submodels of 3 mm threshold and the above, taking $th_1 = 20$ mm as an example.

order to distinguish between 0 and the rainfall above 0.1 mm, we use additionally sigmoid function centered at 0.1 to convert the observed values into labels, as shown in Equation 10:

$$y = \text{sigmoid}(100(R - 0.1)) \quad (10)$$

$R = 0$ corresponds to $y \approx 4.5 \times 10^{-5}$, $R = 0.1$ mm corresponds to $y = 0.5$, and $R = 1$ mm corresponds to $y \approx 1$. Therefore, the label y can guide the submodel of 0.1 mm to exhibit behavior similar to that of a binary classifier of “rainy”/“nonrainy.”

3.2.2. The Loss Function

After constructing the output labels of the model as in Section 3.2.1, a function to measure the disparity between the model's output $\hat{y}$ and the label y is used to optimize the model parameters, which is known as the loss function or objective function, with MAE or MSE loss being the most commonly used variant. However, due to the severe imbalance in the frequency of rainfall of different magnitudes, using the MAE loss function directly, even with a multilevel modeling strategy, can bias the model toward predicting smaller magnitudes that lead to a better average error. Therefore, to balance the impact of samples with different magnitudes on the loss function, we adopt a weighted MAE (wMAE) loss function. First, the errors of all time steps in a forecasting process are summed to obtain the loss of a single forecast:

$$Loss = \sum_{t=1}^T Error_t(\hat{\mathbf{y}}_t, \mathbf{y}_t) \quad (11)$$

$$Error_t(\mathbf{R}_t, \mathbf{y}_t) = \sum_i w_i(i) * |\hat{y}_t(i) - y_t(i)| \quad (12)$$

where $\hat{y}_t(i)$ and $y_t(i)$ are the output and the label at grid point i of time step t , respectively. For submodels of rainfall levels above 3 mm, $y_t(i)$ is obtained from Equation 9. While for the submodel of 0.1 mm, $y_t(i)$ is obtained from

Equation 10. $w_t(i)$ is the weight of grid point i of time step t . The importance of grid points with different observed rainfall values can be adjusted by setting $w_t(i)$. The weight setting method is as follows:

$$w_t(i) = \begin{cases} 1 & , R_t(i) = 0 \\ \left(\frac{\text{mean(PDF)}}{\text{PDF}(R_t(i))} \right)^\alpha & , R_t(i) > 0 \\ 0 & , R_t(i) \text{ is invalid} \end{cases} \quad (13)$$

where $R_t(i)$ is the observed ground truth of rainfall at grid point i of time step t . This means that the weight of a grid point is inversely proportional to the α th power of the probability density of its observed rainfall value, and α takes the value in the range of 0–1. When $\alpha = 0$, all grid points have equal weight. When $\alpha = 1$, the weight is inversely proportional to the probability density of the rainfall value, and the influence of different magnitudes can be perfectly balanced. However, due to interpolation errors in observed values and the extremely low probability density of intense rainfall, using the inverse of probability density as weight directly can lead to computational instability. Therefore, concerning the computational stability, the value of α should not be too close to 1, and the appropriate value of α can be determined with the validation set. When $R_t(i) = 0$, it represents all nonrainy cases, these cases can be simply treated as a uniform distribution, therefore the weight is set to 1.

Particularly, for the submodel of 0.1 mm, the weights for all samples were set to 1, as the model only needs to predict whether there is rainfall or not, and the number of samples with rain and without rain is in the same order of magnitude.

Equation 11 represents the loss for a single forecast, while the total loss of the submodel is the average of all the losses of the forecasts on the training set. After setting up the loss function, the optimal model parameters can be found by optimizing the loss function. In practical, only a batch of samples is selected randomly to calculate the average loss as an estimation of the total loss for each iteration. In this study, the batch size was set between 5 and 10. For deep-learning models, the most commonly used optimization method or parameter update method is stochastic gradient descent and its several variants (Kingma & Ba, 2015).

4. Results and Discussion

In this section, several evaluation methods were used to evaluate the performance of proposed multilevel forecasting strategy and spatial-temporal feature extraction based on GNN. In addition, the characteristics of the models' forecasts in typical rainfall processes were also examined through case studies. The hyperparameters such as L_{hmix} , L_{hprop} , S , and P were determined using the validation set. It was found through several trials that, for submodels of lower levels (0.1 mm, 3 mm), increasing the number of spatial iterations S could lead to better results. However, when S was greater than 5, there was no significant improvement in model performance. And different rainfall thresholds exhibit varying responses to changes of S . For higher rainfall thresholds, smaller S values are preferred. This is possibly due to the fact that intense rainfall events, such as severe convective rainfall, exhibit rapid local variations and shorter durations. Increasing S in these cases may not introduce additional useful information. L_{hmix} and L_{hprop} determines the model's size. If L_{hmix} and L_{hprop} are too small, it will limit the model's representational power. It was found that when L_{hmix} and L_{hprop} were set to be larger than 256, there was no significant improvement in model performance. The hyperparameter P determines the number of f_{prop} , setting multiple f_{prop} can increase the representational power of the model, but it will also make the model training more difficult. Therefore, in all experiments of this paper, L_{hmix} and L_{hprop} were set to be 256, P was set to be 8. For lower rainfall thresholds, suitable S values are around 4 or 5, while for higher rainfall thresholds, appropriate S values are around 2 or 3.

4.1. Evaluation Methods

To evaluate the performance of rainfall forecast, two common verification methods were employed. The threat score (TS) was used as a basic evaluation method, while the Heidke Skill Score (HSS) (Hyvärinen, 2014) was used for rainfall exceeding 10 mm. Additionally, for typical rainfall cases, a spatial verification method called structure-amplitude-location (SAL) was utilized (Wernli, et al., 2008).

TS is one of the common methods for verification of short-term weather forecasting used by meteorological departments. It means the proportion of correctly predicted events (etc. rainfall) among all predicted and/or observed events. The value of TS is between 0 and 1. TS = 1 means a perfect forecast.

Table 2
The Evaluation Results of Different Models Using Point-Based Verification Methods on the Test Set

Method	TS _{0.1}	TS ₃	TS ₁₀	TS ₂₀	HSS ₁₀	HSS ₂₀
GIPMN _{merged}	0.540	0.332	0.184	0.100	0.303	0.179
GIPMN _{0.1}	0.540	—	—	—	—	—
GIPMN ₃	—	0.329	—	—	—	—
GIPMN ₁₀	—	—	0.180	—	0.298	—
GIPMN ₂₀	—	—	—	0.096	—	0.173
GIPMN _{single}	0.515	0.309	0.164	0.081	0.274	0.147
UNet	0.410	0.269	0.143	0.076	0.244	0.141
IFS	0.475	0.279	0.127	0.049	0.219	0.092

Note. The higher score is, the better the performance. The best results are bolded.

HSS is a skill score for categorical forecasts where the accuracy measure is scaled with the reference value from correct forecasts due to chance. The range of the HSS is $-\infty$ to 1. Negative values indicate that the chance forecast is better, 0 means no skill, and a perfect forecast obtains an HSS of 1.

SAL is a spatial verification technique for rainfall forecast. This method first automatically divides the rainfall field in the area into different rainfall objects based on its spatial distribution, and then evaluates the forecast performance from the perspectives of structure (S), intensity (A), and location (L). The ranges of the three values S , A , and L are $[-2, 2]$, $[-2, 2]$, and $[0, 2]$, respectively. S reflects the structural difference between the forecast and observed rainfall fields. $S < 0$ indicates that the forecast area is smaller than the observed area or that the forecast center is larger than the observed center. Otherwise, $S > 0$. A reflects the difference in the average rainfall intensity between the forecast and observed rainfall fields. $A = 0$ indicates that the forecast intensity is consistent with the observed intensity, $A < 0$ indicates that the forecast is weaker than the observed, and $A > 0$ indicates that the forecast is more intense than the observed. L reflects the degree of deviation between the centroid of the forecast rainfall and that of the observed rainfall. The smaller the L value, the closer the forecast and observed rainfall areas.

4.2. Overall Evaluation

This section mainly verifies the effectiveness of the GIPMN model and the multilevel merging strategy proposed in this paper on the test set using the point-based verification methods mentioned above. To verify the effectiveness of multilevel modeling, a GIPMN model that directly outputs the entire forecast rainfall field was used for comparison, which we refer to as “GIPMN_{single}.” To evaluate the ability of GIPMN as a feature extractor in extracting multiscale features, we used a deep-learning benchmark model UNet (Ronneberger, et al., 2015). The UNet architecture is named after its U-shaped structure, which expands the “perceptual field” via convolution and down-sampling in a “down” process, followed by the gradual recovery of feature maps to their original size via the “up” process of deconvolution and up-sampling. Skip connections are included between feature maps at the same levels of these two processes. Therefore, UNet uses multiple convolutions and deconvolutions to fuse multiscale features. UNet has been used for correction of numerical forecast products and precipitation forecast (Grönquist, et al., 2021; Han et al., 2021; Hess & Boers, 2022), and it is representative enough as a benchmark model. GIPMN_{single} and UNet used the predictors in both Table 1 (a and b) as inputs. The input of UNet is the concatenation of the predictors of all forecast time steps sequentially, so the number of input channels of the model is the number of forecast factors times the number of forecast time steps, and the output is the forecasted rainfall field of all corresponding time steps. These two models were also trained using the loss function defined by Equations 11–13, but since they both directly output the value of rainfall, their true label and multilevel models used were different, and they were obtained by a simple scaling:

$$y = R/R_0 \quad (14)$$

In this paper, $R_0 = 10$ mm. The output was multiplied by R_0 to get the forecasted rainfall value.

It is worth noting that the IFS ensemble forecast is statistically more reliable than the HRES, and can offer valuable uncertainty information for the forecast. However, the proposed method relies on the deterministic output variables of the HRES with a higher spatial resolution and doesn't take into account the initiation error distribution. Additionally, the rainfall thresholds chosen in IFS ensemble for probabilistic forecasting is different with ours. For these reasons, the IFS ensemble forecast product was not chosen as a baseline in this paper.

The evaluation results of different models using TS and HSS on the test set are shown in Table 2. To evaluate the performance, all valid grid points at all lead times of all forecasts in the test set were included in the verification function and the calculation was performed only once. Therefore, the score represents the statistical result of the entire test set.

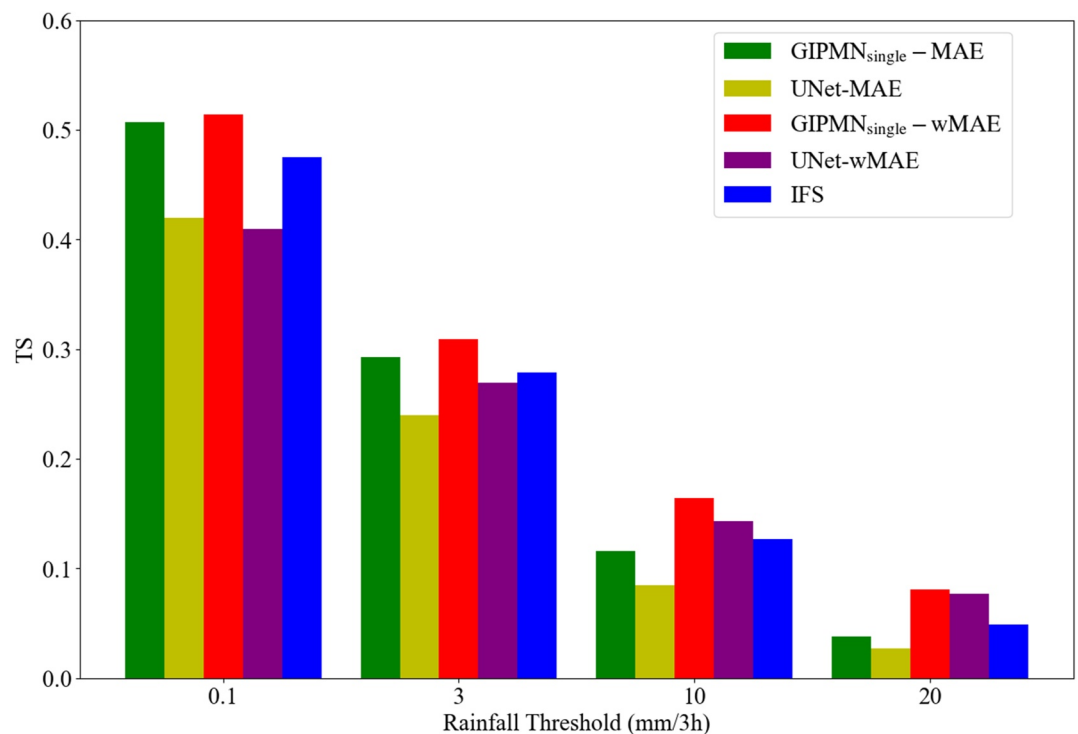


Figure 8. Comparison of threat score (TS) between models trained MAE loss and weighted MAE (wMAE) loss at each rainfall threshold.

From the results in Table 2, it can be seen that both GIPMN and UNet produced positive skills relative to IFS on cases of rainfall above 10 mm, and the score of GIPMN_{single} was higher than UNet, which verifies the effectiveness of the proposed spatiotemporal feature extracting mechanism. However, UNet produced negative skill on cases below 10 mm, even though it demonstrated positive skill on the validation set (not shown). This indicates that GIPMN has stronger generalization ability than UNet. Furthermore, after dividing the models into multiple thresholds, the score of each threshold was improved compared to the single model GIPMN_{single}. Moreover, the merged model GIPMN_{merged} after multilevel merging had slightly higher scores than the submodels of the individual thresholds, indicating the effectiveness of the merging strategy. In terms of TS, GIPMN_{merged} showed significant improvements over IFS for each rainfall threshold. The improvement was even more significant for rainfall thresholds above 10 mm, with the TS and HSS of 10 mm improving by 45% and 38% over IFS, respectively, and the TS and HSS of 20 mm improving by 104% and 95%, respectively. Figure 8 shows a comparison of TS for models trained using MAE loss and the proposed wMAE loss. It can be seen that the wMAE loss improved TS scores for GIPMN_{single} at all thresholds. And it also improved TS scores for UNet at thresholds above 3 mm. GIPMN_{single} trained using MAE loss showed negative skills relative to IFS for threshold above 10 mm, while the UNet trained using MAE loss showed negative skills for all thresholds. This indicates that the proposed wMAE loss function can effectively balance the influence of samples with different rainfall intensities. Figure 9 shows the TS of each model at different forecast lead times. It can be seen that GIPMN performed the best in almost all forecast lead times. And the performance of GIPMN was more stable against lead time than that of UNet. IFS showed a periodic-like change in TS for the 0.1, 10, and 20 mm thresholds, which may be related to the assimilation method of the ECMWF models (Bauer, et al., 2015).

4.3. Case Study

Based on the SAL spatial verification method, the application results of GIPMN and UNet on two rainfall cases are discussed, respectively. Due to the existence of invalid grid points with missing observed value in the verification area, the observed values of the invalid grid points were set to 0 to avoid numerical errors. If there are too many invalid grid points, SAL will be not credible.

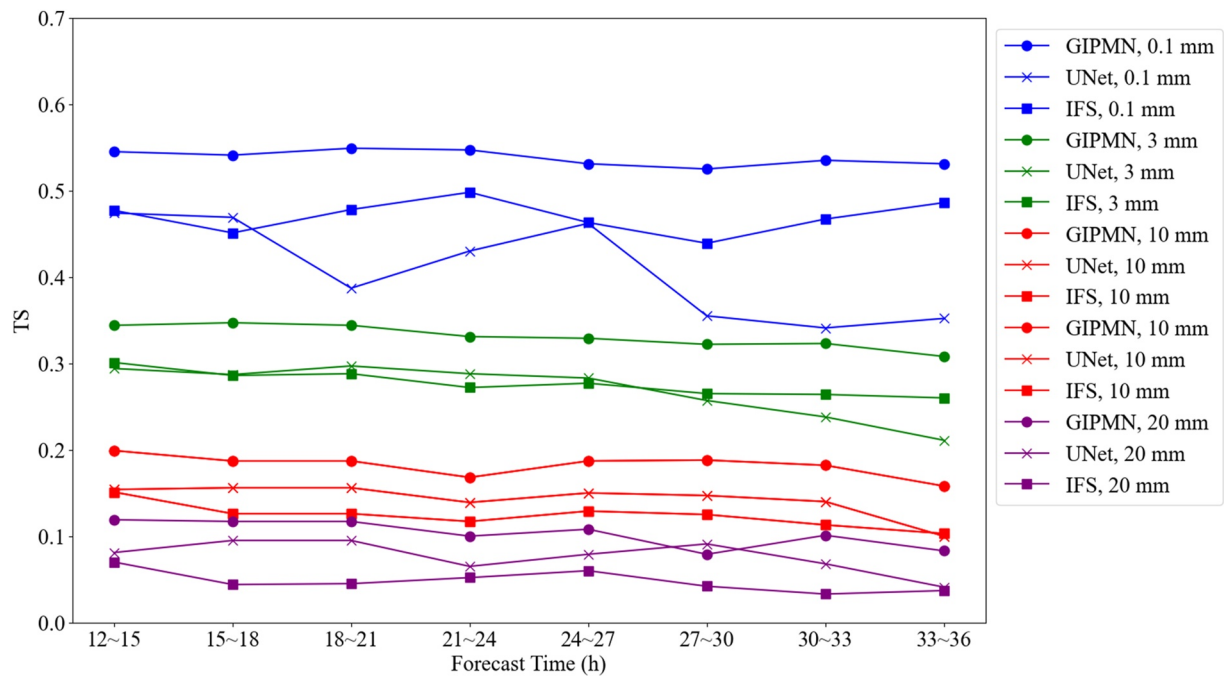


Figure 9. Comparison of threat score (TS) between different models of each forecast time step.

Figure 10 illustrates the performance in Severe Typhoon “Muifa,” which was generated over the Northwestern Pacific Ocean at 00:00 UTC 8 September 2022 and landed in Zhoushan, Zhejiang on 12:30 UTC 14 September. It can be seen that the forecasts of IFS and GIPMN performed similarly in the first three time steps, but in forecast lead time of 21–36 hr, the position of heavy rain forecasted by IFS deviated westward, as illustrated by the red circle marker. The GIPMN predicted more accurate positions of the heavy rainfall areas. The structural similarity to the observation of UNet forecast is lowest of the three models. Moreover, UNet also outputted false rainfall at the edges of the output area. This is because the fully convolution structure adopted by UNet matches the convolution kernel by replicating the feature map on the edge, which may cause the signal after convolution to be amplified. Furthermore, the comparison of the SAL results of all models at timesteps 5, 7, and 8 also demonstrates that GIPMN performed the best most of the time for S and A (Figure 11). The L scores of the three models are similar and all below 0.2, indicating accurate prediction of the locations of the main rainfall areas. The SAL at timestep 6 is unavailable due to missing observation of the main rainfall area. The 10 and 20 mm TS scores of GIPMN in this case are 0.441 and 0.476, respectively, which are also much higher than 0.308 and 0.187 of IFS.

Figures 12 and 13 show the performance in a convective rainstorm located in the Yangtze River Valley, which happened under the influence of cold air and a strong shear line, with divergence reaching $-4 \times 10^{-5} \text{ s}^{-1}$ on the 850 hPa weather chart (not shown) and a maximum rainfall above 30.0 mm/hr. Compared with the observation, both IFS and UNet underestimated the rainstorm, while GIPMN has a good performance in illustrating the whole rainstorm process, especially after lead time of 21 hr. However, GIPMN overestimated the sizes of intense rainfall centers in this case. From the SAL scores, IFS had the best S scores most of the time, indicating better prediction of the proportion of rainfall center area size. However, S cannot evaluate the accuracy of the predicted rainfall center location. While GIPMN had the best A scores most of the time, indicating better predictions of total rainfall amount. The L scores of the three models are also similar and all below 0.1, indicating accurate prediction of the locations of the main rainfall areas. The 10 and 20 mm TS scores of GIPMN in this case are 0.327 and 0.156, respectively, which are also higher than 0.256 and 0.088 of IFS.

5. Summary

This study proposes a structured graph neural network model referred to as GIPMN, to improve NWP rainfall forecasting. The model extracts and fuses multiscale spatial-temporal features of multipredictors by designing two different grid information propagation and fusion functions. To enable the model better at handling different

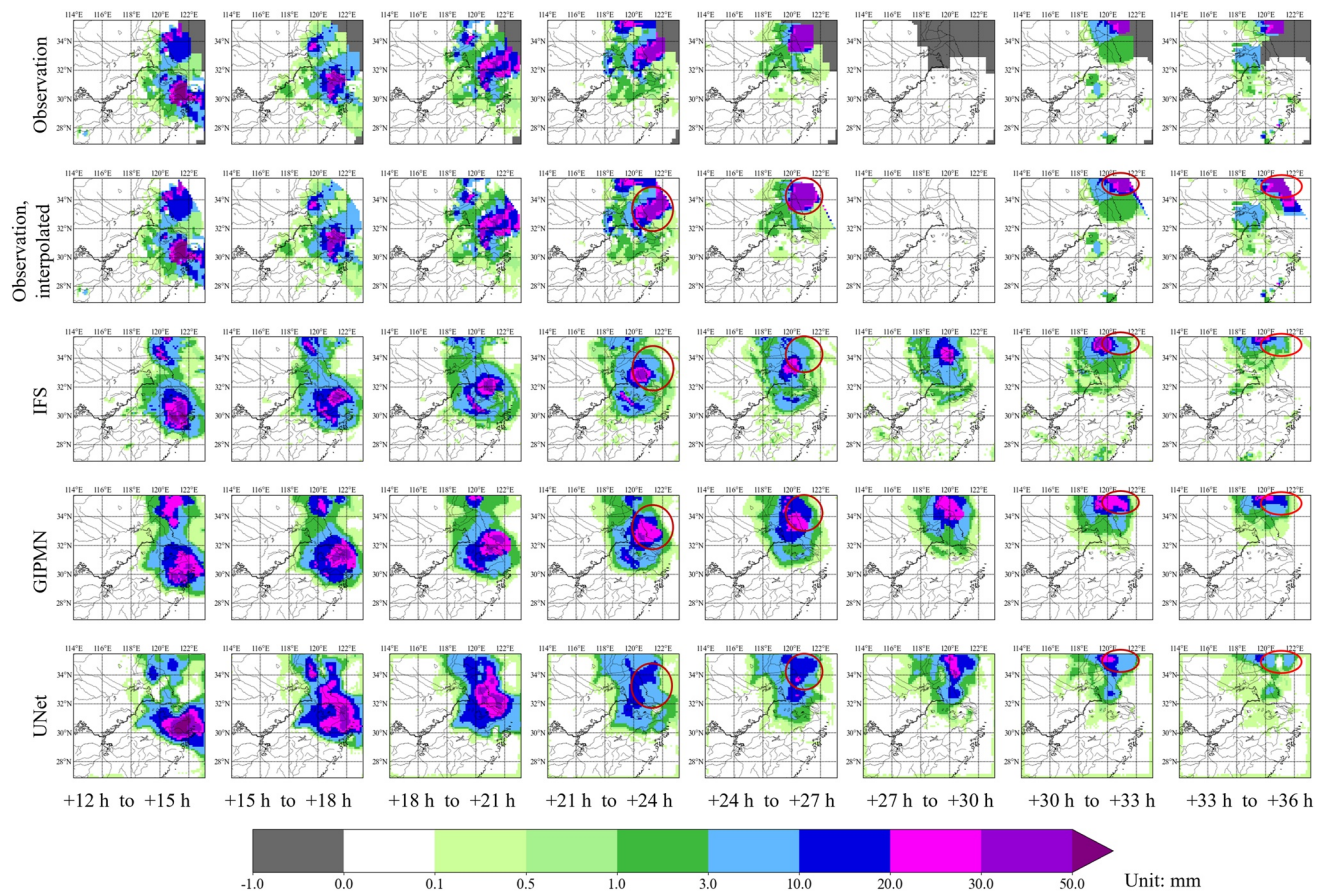


Figure 10. Forecasts of rainfall fields from 00:00 UTC 14 September to 24:00 UTC 14 September 2022 by different models. The grid points of the observation rainfall field with nonavailable data are displayed in gray. The interpolated observation field shown in the second row was obtained by linear interpolation within the convex hull enclosed by the grid points with valid observations, hence the values outside the convex hull are still missing.

types of rainfall, a multilevel modeling strategy is adopted. For the submodel of each threshold, a soft classification label that reflects the probability of rainfall above that threshold is designed, allowing the model to learn the probability of rainfall above each threshold. And a weighted MAE loss is proposed to balance the impact of samples with different rainfall intensities. This multilevel forecasting strategy is applicable to any data-driven model.

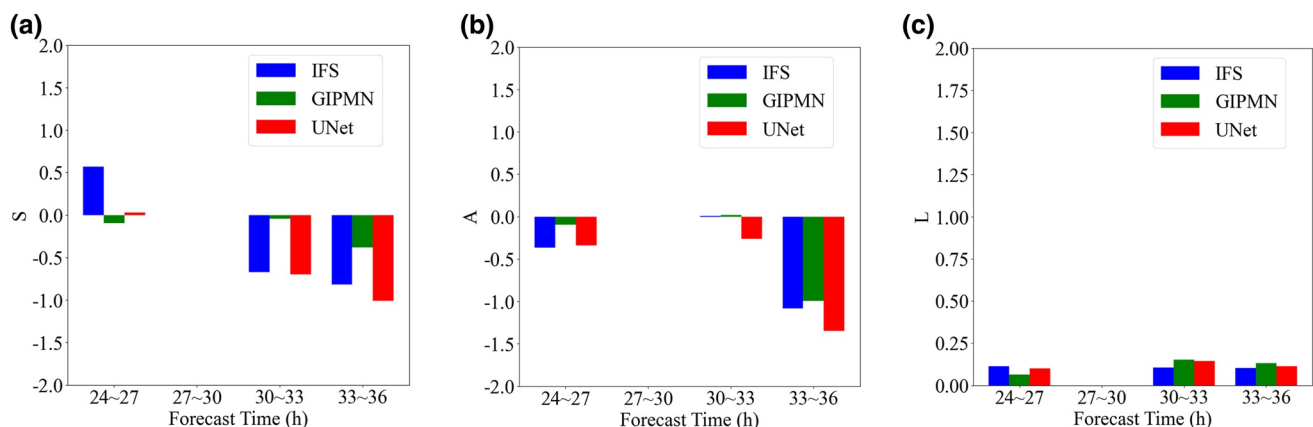


Figure 11. Structure-amplitude-location (SAL) scores of forecasts of different models from 12:00 UTC 14 September to 24:00 UTC 14 September 2022. The closer the score is to 0, the better the performance.

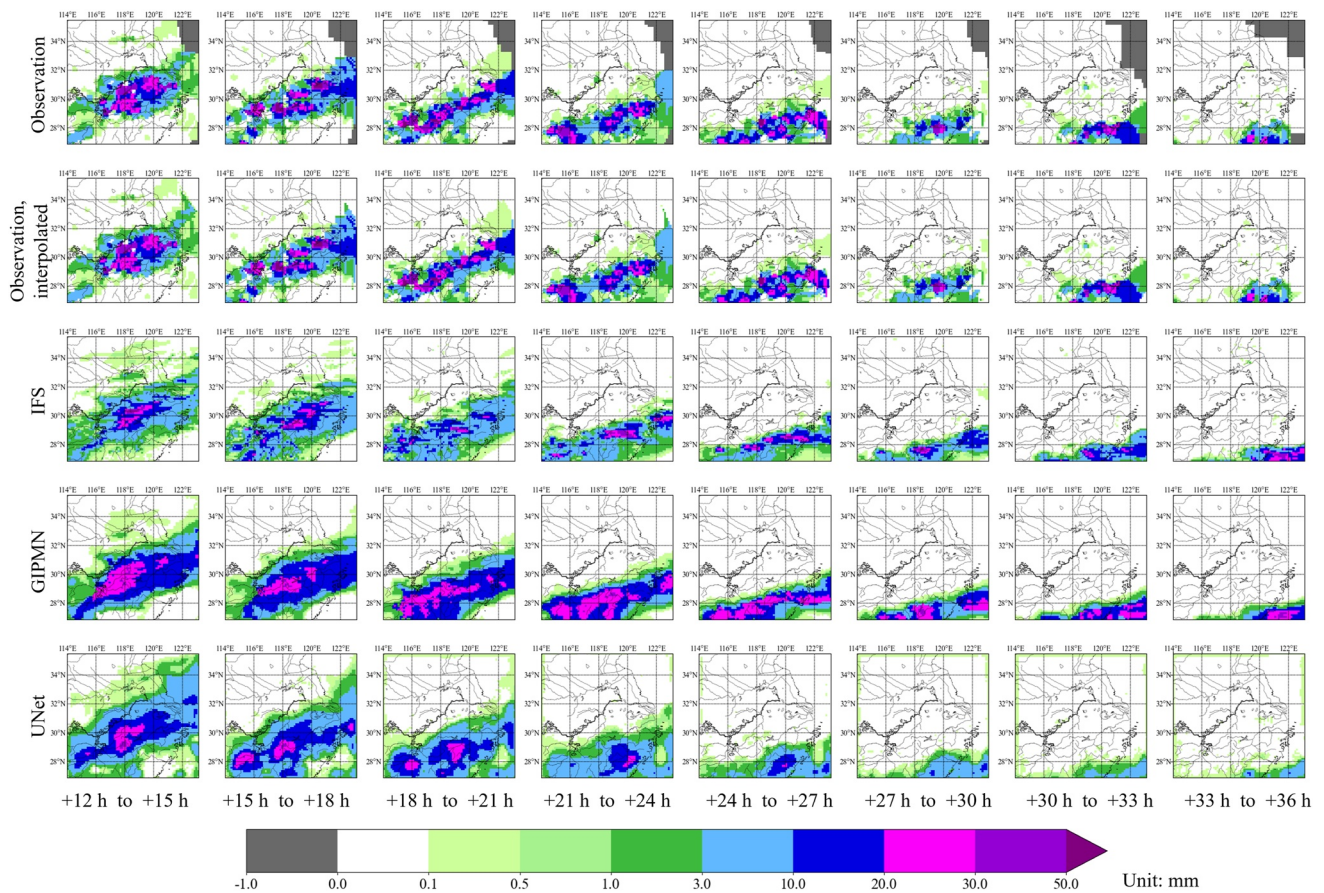


Figure 12. Same as Figure 10, but for 00:00 UTC 4 June to 00:00 UTC 5 June 2022.

The proposed model was applied to the deterministic forecast products of IFS from ECMWF for the central part of East China. The basic verification results on the test set show that GIPMN significantly outperforms IFS in forecast skills for all thresholds, especially for thresholds above 10 mm. The comparison between the model trained with MAE loss and the model trained with wMAE loss shows that wMAE loss effectively balances the contribution of samples with different rainfall amount to the total loss, making it easier for the model to learn the features of samples with intense rainfall. In addition, the skills of each threshold have also improved to some extent after adopting the multilevel forecasting strategy, which also validates the effectiveness of the proposed multilevel modeling strategy.

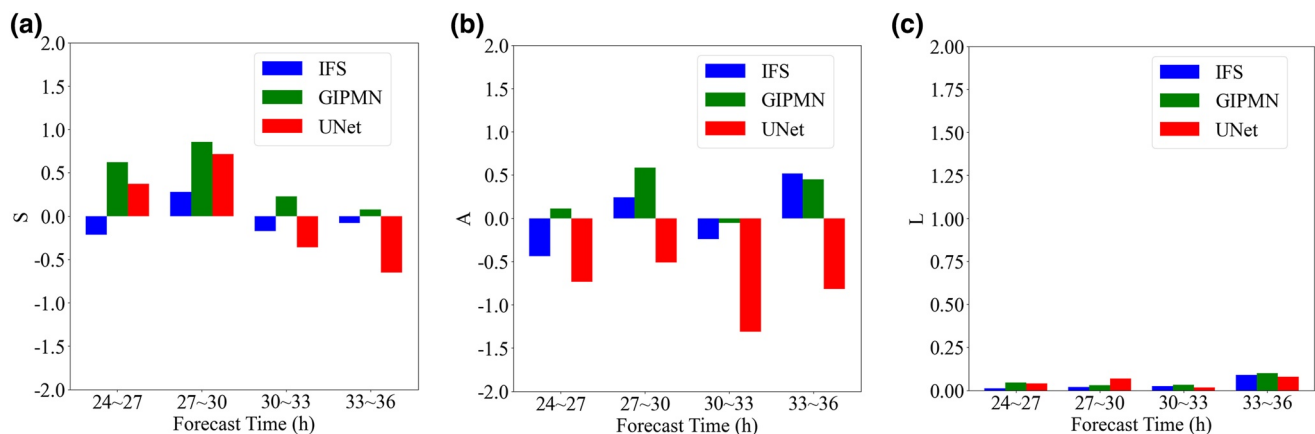


Figure 13. Same as Figure 11, but for 00:00 UTC 4 June to 00:00 UTC 5 June 2022.

A case study on two typical precipitation events was performed with a spatial verification method, SAL. The results showed that compared with the other two models, GIPMN more accurately predicted the location of the precipitation central area and the intensity of the precipitation center, indicating that GIPMN can learn the spatial distribution of precipitation. However, GIPMN's forecast may tend to overestimate the size of the central area of intense rainfall.

The uncertainty of NWP precipitation forecast comes not only from initiation errors and model approximation errors, but also from small-scale processes that the model cannot resolve, such as local convection. Accurately measuring the uncertainty is particularly important for extreme weather forecasting. NWP usually deals with uncertainty with ensemble forecast, but at the cost of sacrificing model resolution and incurring a larger computational burden. A future research direction is to utilize probabilistic reasoning such as Bayesian inference to model the uncertainty of small-scale processes, and to use high-precision observation data to guide the model to learn this uncertainty. Through an accurate description of uncertainty and variability, decision-makers can be more informative to make more reasonable decisions.

Data Availability Statements

The code and data set used in this paper is publicly available on Zenodo (Peng, 2023).

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